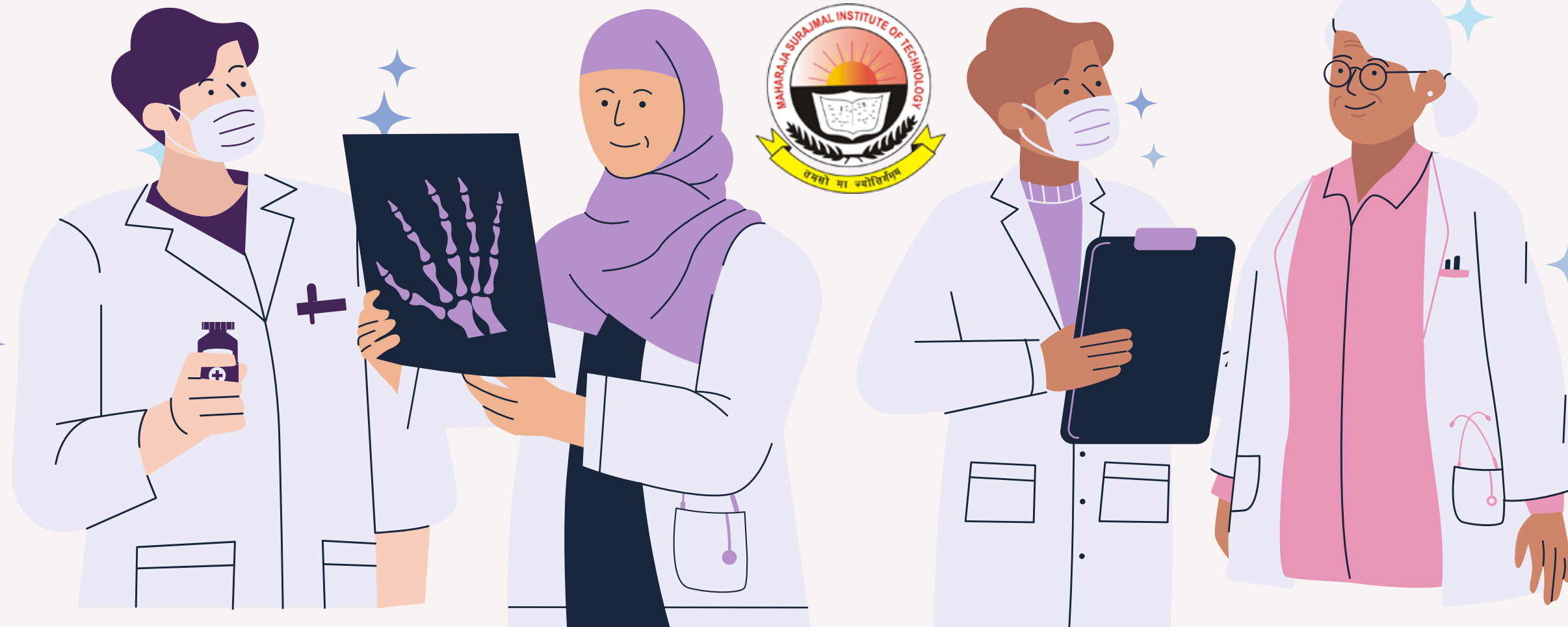


First Progress Presentation

Advanced Deep Learning Framework for Multimodal Melanoma Detection in Human Subjects

Under the supervision of
Dr. Neeti Sangwan



Aditya Gupta (00396302721), Avinash Thakur (02796302721), Manya Bora (35496302721)

Table of Contents

1	Introduction.....	3	7	CNNs.....	9
2	Prproblem Statement.....	4	8	Our approach.....	10
3	Features.....	5	9	Tech stack	11
4	Why this topic	6	10	Data set and processing...	12
5	What is Melanoma.....	7	11	Potential Impact.....	13
6	Types of Melanoma.....	8	12	Conclusion.....	14



Introduction

Melanoma, which means "black tumor," is the most dangerous type of skin cancer.

Melanoma incidence has dramatically increased over the past 30 years.

It's widely accepted that increasing levels of ultraviolet exposure are one of the main reasons for this rapid rise in the number of melanoma cases.

Early detection and response are crucial to improving skin cancer therapy effectiveness and reducing mortality.

Dermatologists use dermoscopy, ocular inspection, and professional expertise to discover skin problems.

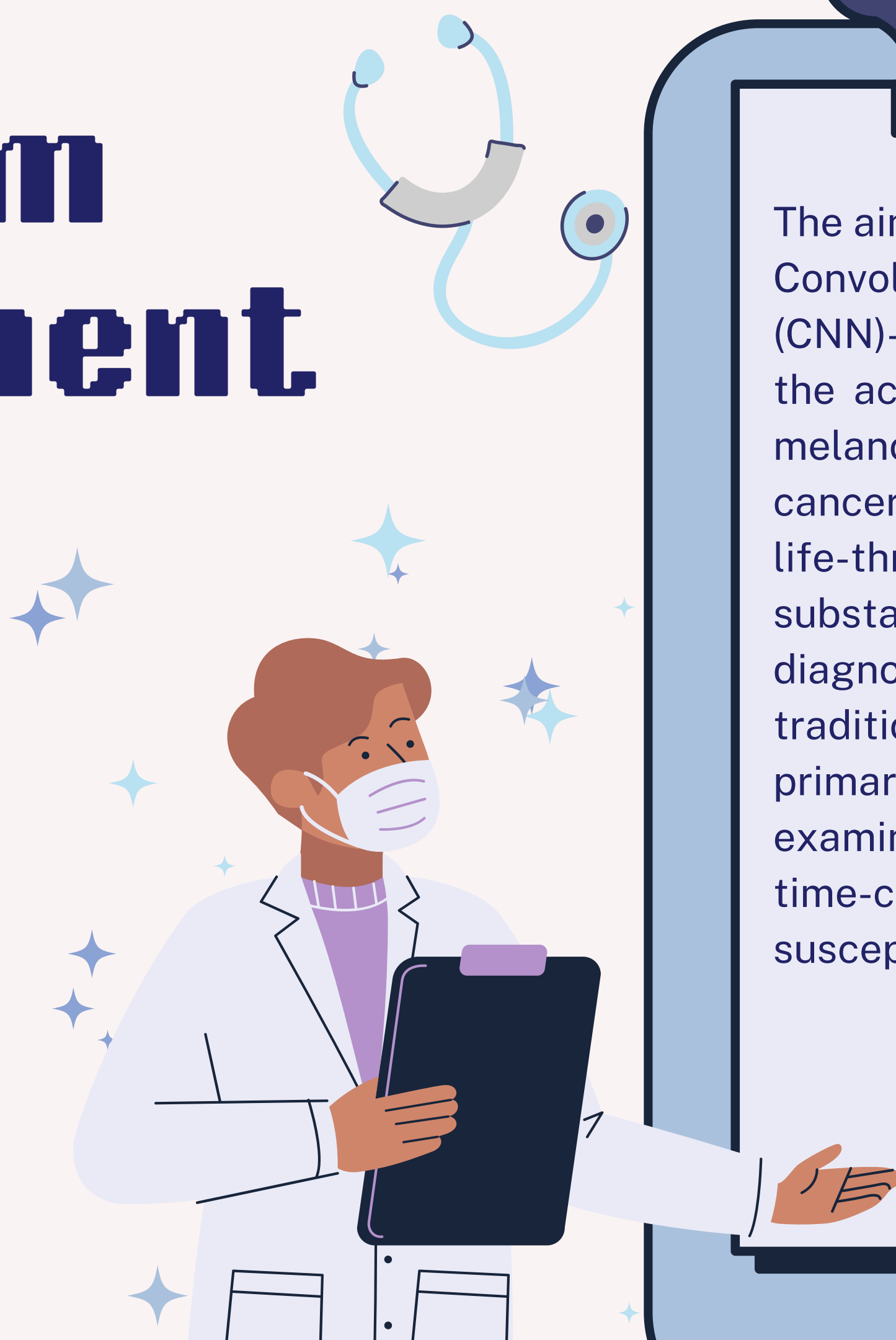


Depending on the clinician's experience, this method may be subject to time-consuming, and error-prone.



Problem Statement

An exploration of innovations in medicine via machine learning



The aim of this project is to develop a Convolutional Neural Network (CNN)-based deep learning model for the accurate and early detection of melanoma from medical images. Skin cancer is a prevalent and potentially life-threatening disease, with a substantial number of cases diagnosed globally each year. The traditional methods of diagnosis, primarily reliant on visual examination by dermatologists, are time-consuming, subjective, and susceptible to errors.

Exploring Breakthroughs and Challenges

Accurately Detecting lesion

Develop a model capable of differentiating between benign and malignant melanomas with a high degree of accuracy.

Enable Early Diagnosis

Facilitate the early detection of melanoma, which is crucial for timely intervention and improved patient outcomes.

Reduce Subjectivity

Mitigate the subjectivity inherent in human visual assessments and provide a consistent and objective evaluation of skin lesions.

Enhance Accessibility

Create a user-friendly and accessible solution that can be deployed via a mobile application, enabling individuals to perform initial assessments at home.

Why this topic?

Melanoma, which means "black tumor," is the most dangerous type of skin cancer. It grows quickly and has the ability to spread to any organ.

Its early detection is pivotal in ensuring effective treatment and reducing mortality rates as treatment success is directly related to the depth of the cancerous growth.

The rapid advancements in deep learning and computer vision technologies have opened up unprecedented opportunities to tackle complex medical challenges, such as skin cancer diagnosis.



What is a Melanoma



Melanoma is a kind of skin cancer that starts in the melanocytes. Melanocytes are cells that make the pigment that gives skin its color. The pigment is called melanin.

It's the most invasive skin cancer with the highest risk of death. While it's a serious skin cancer, it's highly curable if caught early.

Types of Melanoma

Superficial spreading melanoma

Superficial spreading melanoma tends to start growing across the skin before growing deeper down into it. It can develop anywhere on the body.

Lentigo maligna melanoma

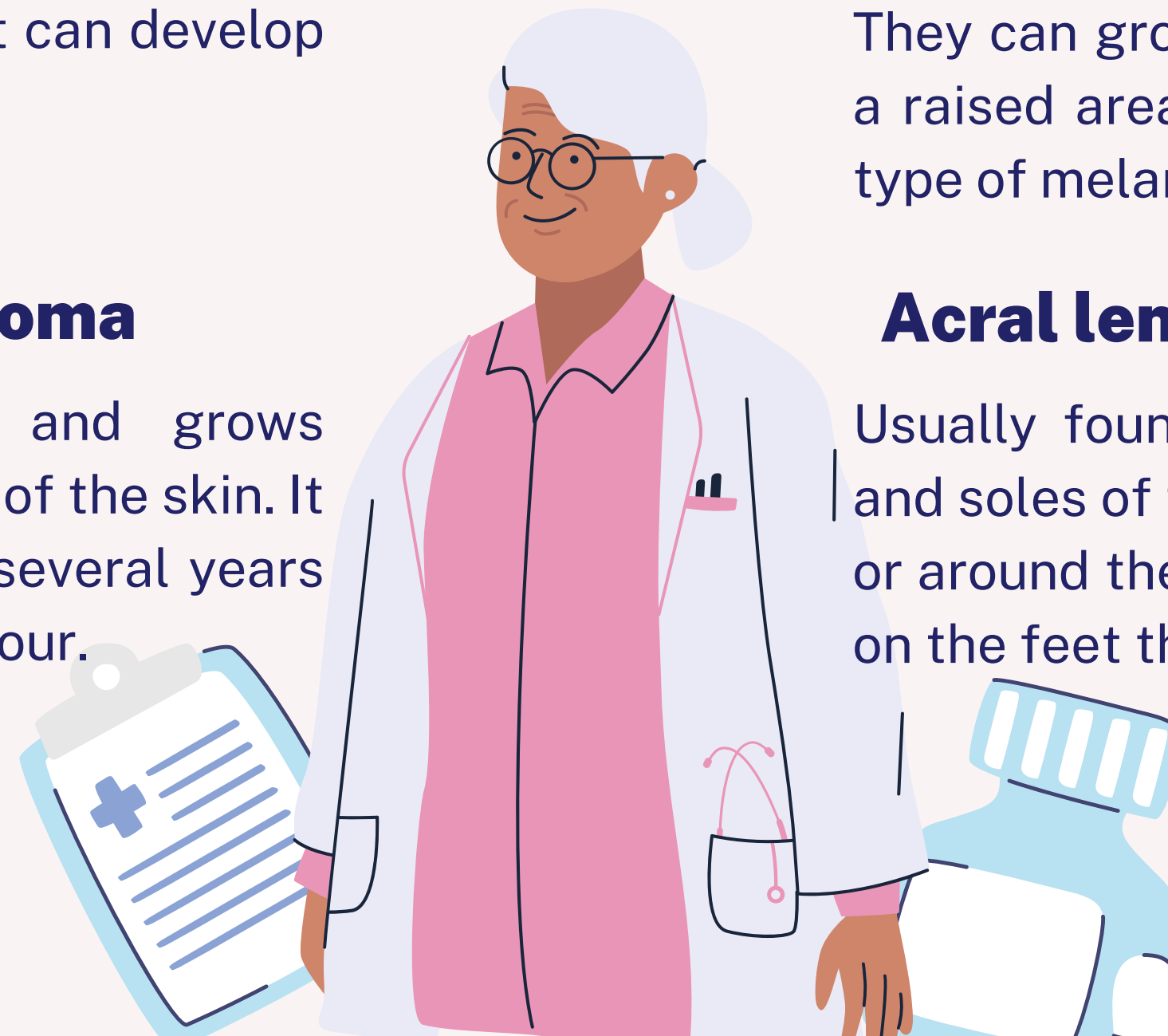
A lentigo maligna is flat and grows outwards in the surface layers of the skin. It might slowly get bigger over several years and might change shape or colour.

Nodular melanoma

Nodular melanomas tend to grow downwards into the deeper layers of skin. They can grow quite quickly. There is often a raised area on the skin surface with this type of melanoma.

Acral lentiginous melanoma

Usually found on the palms of the hands and soles of the feet. It can also grow under or around the nails. It's much more common on the feet than on the hands.



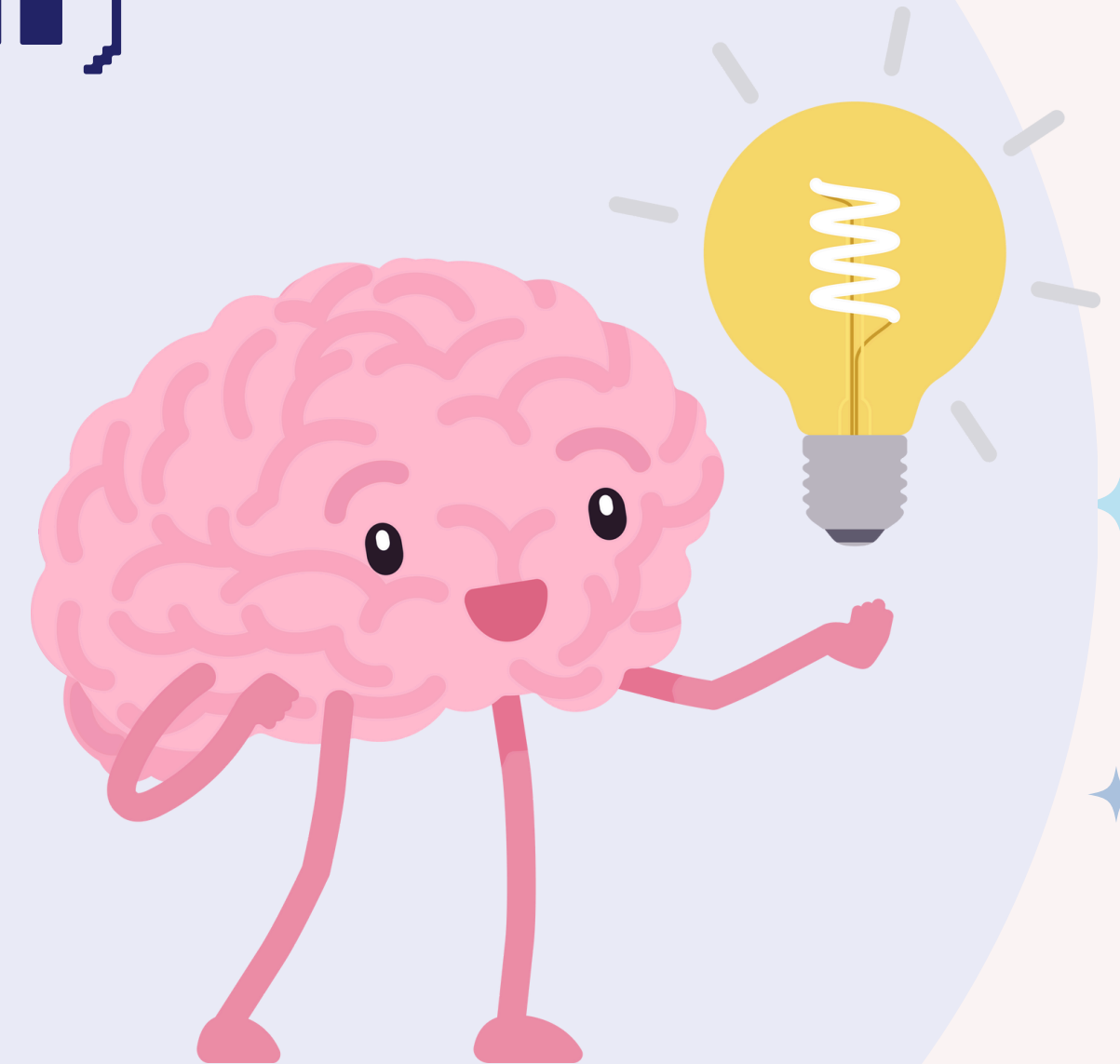
CONVOLUTIONAL NEURAL NETWORK (CNN)

Definition

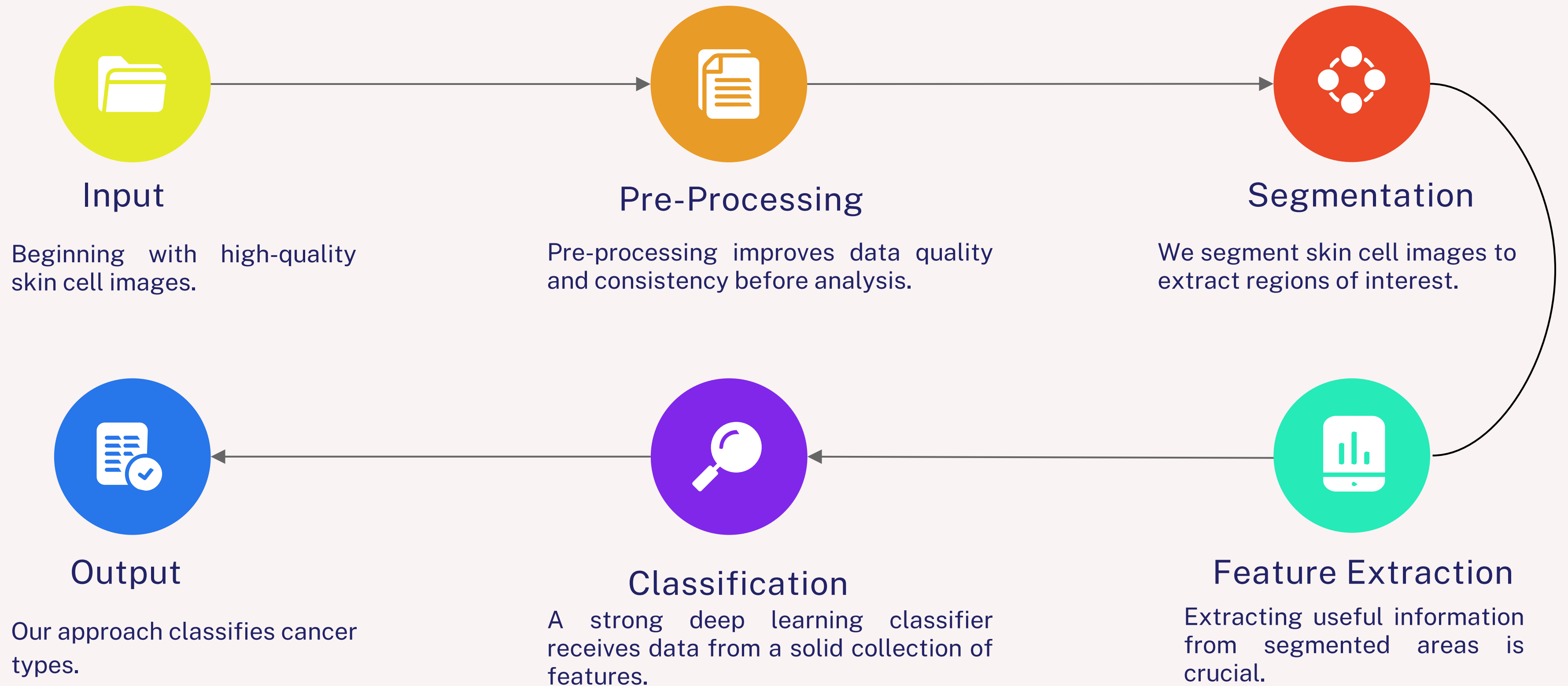
CNNs are a powerful tool for machine learning, especially in tasks related to computer vision. They are a specialized class of neural networks designed to effectively process grid-like data, such as images.

Architecture

The architecture of CNNs is inspired by the visual processing in the human brain. It's key components are Convolutional layers, Pooling Layers, Activation Functions and Fully Connected Layers.



OUR APPROACH



Technology Stack

PYTHON

01

KERAS

04

TENSORFLOW

02

MOBILE APP
DEVELOPMENT TOOLS

05

SCIKIT-LEARN

03

GOOGLE COLLAB

06



Data set and Image processing

Dataset Overview

Our experiment uses a dataset of 10,000 skin cell pictures.

Significance

This dataset helps us train and test our deep learning model.

Diversity & Realism

Our collection includes photos of a variety of skin types, conditions, and ages, representing clinical practice.

Metadata Utilisation

Each picture has clinical metadata with diagnostic labels and patient characteristics.

Image Preprocessing

We preprocess photos to standardize and improve quality before analysis

Potential impact of this project

Improved Accuracy

The deep learning model has the potential to surpass human-level accuracy in detecting skin cancer, false reducing the chances of negatives and positives.

Accessibility

Making this technology accessible via a mobile application can reach a broader population, especially in underserved areas with limited access to healthcare resources.

Enhanced Healthcare Efficacy

The project can contribute to the overall efficiency of healthcare systems by streamlining the diagnostic process, enabling faster and more accurate deductions.

Conclusion

In this project, we developed a deep learning model to detect melanoma lesions employing a large dataset of dermoscopic images.

The results indicate that deep learning can serve as a powerful tool in aiding detection of melanoma early, potentially improving patient outcomes through timely intervention.



Thank you

